Weak Normalization of the Simply Typed λ-Calculus

Following Girard's *Proofs and Types*, Chapter 4

The Simply Typed λ-Calculus

Types and Terms

Types

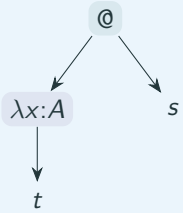
$$A, B ::= \iota \mid A \to B$$

where ι ranges over base types.

Terms

$$t, s ::= x \mid \lambda x{:}A.\, t \mid t\ s$$

Terms as Trees



A term is a finite tree.

Two subtrees are either **disjoint** or one **contains** the other.

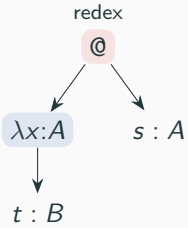
Typing Rules

$$\frac{}{\Gamma, x : A \vdash x : A}(\text{Var}) \qquad \frac{\Gamma, x : A \vdash t : B}{\Gamma \vdash \lambda x{:}A.\, t \ : \ A \to B}(\text{Abs})$$

$$\frac{\Gamma \vdash t : A \to B \qquad \Gamma \vdash s : A}{\Gamma \vdash t\ s : B}(\text{App})$$

β-Reduction

A **redex** is a subterm of the form $(\lambda x{:}A.\, t)\, s$. Its **contractum** is $t[x := s]$.



$\longrightarrow_\beta$

$t[x := s] : B$

One-step reduction $t \to t'$: replace one redex in t by its contractum.

Normal Forms

A term t is in **normal form** if it contains no redex.

A term t is **weakly normalizing** if there exists a finite reduction sequence

$$t \to t_1 \to \cdots \to t_n$$

where t_n is in normal form.

Main Results

Weak Normalization Theorem

Theorem (Weak Normalization)

Every well-typed term in the simply typed λ-calculus has a normal form.

That is: if $\Gamma \vdash t : A$, then there exists a reduction sequence $t \to^* t'$ where t' is in normal form.

The proof provides an explicit **reduction strategy** that terminates:

1. Assign a measure $\mu(t) = (n, m)$ to each term.
2. Show that a suitable choice of redex strictly decreases μ in the lexicographic order.
3. Conclude by well-foundedness of $(\mathbb{N} \times \mathbb{N}, <_{\text{lex}})$.

Consequence: Decidability of Intuitionistic Logic

Corollary

Provability in the intuitionistic propositional calculus (IPC) is decidable.

By the Curry–Howard correspondence, proofs of A in IPC correspond to closed terms of type A in STLC, and cut-elimination corresponds to normalization.

The subformula property. By weak normalization, if A is provable then it has a *normal* (cut-free) proof. In a cut-free proof, there are no detours: as you go up the proof tree, the formulas appearing in each step are *subformulas* of the formulas below. The steps get shorter as you ascend.

Decidability (continued)

Why this gives decidability. Since the formulas get strictly shorter as you move up a cut-free proof, and there are only finitely many subformulas of A and Γ, both the depth and the branching of any cut-free proof are bounded. There are therefore only finitely many possible cut-free proofs of A from Γ.

So: enumerate all candidate cut-free proofs and check whether any one is valid. If one is found, A is provable; if none works, A is not provable. $\square$

The Degree Measure

Degree of a Type

Definition

The **degree** of a type (ι is an atomic type):

$$\partial(\iota) = 0 \qquad \partial(A \to B) = \max(\partial(A), \partial(B)) + 1$$

Examples:

$$\partial(\iota \to \iota) = \max(0, 0) + 1 = 1$$
$$\partial((\iota \to \iota) \to \iota) = \max(1, 0) + 1 = 2$$
$$\partial(\iota \to (\iota \to \iota)) = \max(0, 1) + 1 = 2$$

Key property: For $A \to B$, both $\partial(A) < \partial(A \to B)$ and $\partial(B) < \partial(A \to B)$.

This is immediate: $\partial(A) \leq \max(\partial(A), \partial(B)) < \max(\partial(A), \partial(B)) + 1 = \partial(A \to B)$.

Degree of a Redex

Definition

The **degree of a redex** $r = (\lambda x{:}A.\, t)\, s$, where $t : B$, is:

$$\partial(r) = \partial(A \to B) = \max(\partial(A), \partial(B)) + 1$$

Observation

If r is a redex of type T, then $\partial(r) > \partial(T)$.

Proof: $r = (\lambda x{:}A.\, t)\, s$ has type $T = B$. Then
$\partial(r) = \partial(A \to B) \geq \partial(B) + 1 > \partial(B) = \partial(T)$.

Degree of a Term

Definition

The **degree of a term** is:

$$d(t) = \max\{\partial(r) : r \text{ is a redex occurring in } t\}$$

Convention: $d(t) = 0$ if t is in normal form.

The Measure $\mu(t)$

Definition

For a term t, define:

$$\mu(t) = (d(t),\ \nu(t)) \in \mathbb{N} \times \mathbb{N}$$

where $\nu(t)$ is the number of redexes of degree $d(t)$ in t.

Order: **lexicographic** on $\mathbb{N} \times \mathbb{N}$.

$$(n', m') < (n, m) \iff n' < n \ \text{ or } \ (n' = n \text{ and } m' < m)$$

This is a well-order on $\mathbb{N}^2$, so any strictly decreasing sequence is finite.

Normal form corresponds to $\mu(t) = (0, 0)$.

Statement of Lemmas

The Three Lemmas

All three lemmas concern what happens to redexes under substitution and reduction.

Lemma 1 (Substitution Lemma)

If $x : U$ then $d(t[x := u]) \leq \max(d(t), d(u), \partial(U))$.

Lemma 2 (Degree Non-Increase)

If $t \to u$ (one-step β-reduction), then $d(u) \leq d(t)$.

Lemma 3 (Key Lemma)

Let r be a redex of maximal degree n in t, with all redexes strictly inside r having degree $< n$. Then contracting r strictly reduces the count of degree-n redexes.

Subject Reduction (Lemma 0)

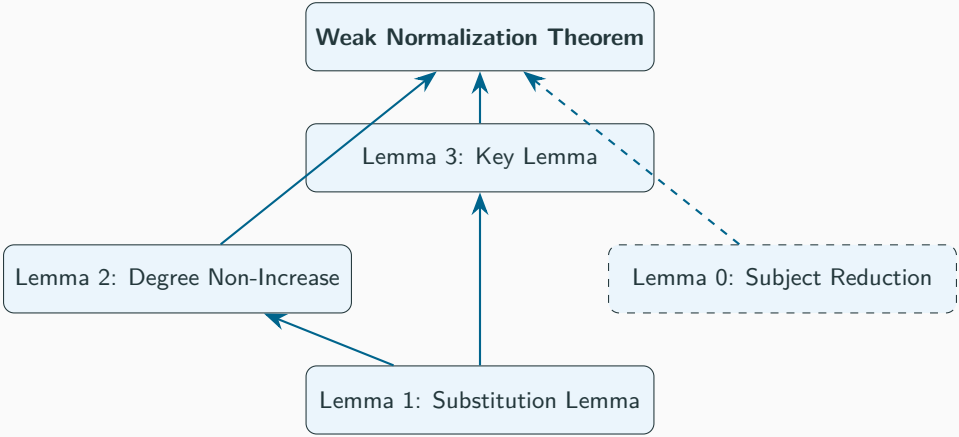
Lemma 0 (Subject Reduction / Type Preservation)

If $\Gamma \vdash t : A$ and $t \to_\beta t'$, then $\Gamma \vdash t' : A$.

This is the "hidden" lemma: we use it throughout (contracting a redex preserves the type), and it is intuitively obvious—reducing a well-typed term should not break its type.

The proof is a straightforward induction on the typing derivation; the only interesting case is when t is itself a redex. It is left as a **homework exercise**.

Dependency Structure



{We prove the theorem first (assuming the lemmas), then Lemma 1 as a warm-up on substitution, then Lemma 2, and finally Lemma 3. Lemma 0 is used implicitly throughout and left as homework.}

Proof of the Theorem

Reduction Strategy

Goal: Given t with $\mu(t) = (n, m)$, find a redex whose contraction yields t' with $\mu(t') < (n, m)$.

Strategy:

1. Let $n = d(t)$. If $n = 0$, then t is normal. Done.
2. Among all degree-n redexes in t, choose one that is innermost: it contains no other degree-n redex. Contract it.

Such a redex always exists: since terms are finite trees, we can traverse inward from any degree-n redex until we find one that contains no other degree-n redex. This innermost redex satisfies Lemma 3's hypothesis by construction.

Proof of Weak Normalization

Proof. By well-founded induction on $\mu(t) = (d(t), \nu(t)) \in (\mathbb{N}^2, <_{\text{lex}})$.

Base case: $d(t) = 0$ means t has no redexes — already in normal form.

Inductive step: Let $n = d(t) > 0$. Contract an innermost degree-n redex r to get t'.

- **Lemma 2** $\Rightarrow$ $d(t') \leq n$.
- **Lemma 3** $\Rightarrow$ t' has strictly fewer degree-n redexes.

Either $d(t') < n$ (first component drops) or $d(t') = n$ and $\nu(t') < \nu(t)$ (second component drops). Either way $\mu(t') < \mu(t)$, so by IH t' has a normal form. $\square$

Warm-up: What Does Substitution Do to Redexes?

The Key Question

Before proving the lemmas, let us build intuition for the central issue in the entire proof: when we perform a substitution $t[x := u]$, what happens to redexes?

Lemma 1 (restated)

If $x : U$ then
$d(t[x := u]) \leq \max(d(t), d(u), \partial(U))$.

Substitution replaces every occurrence of x in t by a copy of u. So a redex ρ' in the result is either:

1. an old redex of t that survived the substitution,
2. a copy of a redex that was already in u, or
3. a genuinely new redex, created by the interaction between t and u.

Source 1: Old Redexes of t

If $(\lambda y{:}C.\, P)\, Q$ was a redex in t, with $P : B$, it becomes $(\lambda y{:}C.\, P[x := u])\, Q[x := u]$ in $t[x := u]$. The type annotation on the λ is still C, and the body still has type B (by Subject Reduction), so the degree of this redex is unchanged. Bounded by $d(t)$.

E.g. $t = (\lambda y{:}\iota.\, x)\, a$, $u = b$. The redex $(\lambda y{:}\iota.\, b)\, a$ has the same type $\iota \to \iota$, degree 1.

Warm-up (Lemma 1)

Bound $d(t[x := u])$

Hyp: $x : U$, substituting u for x in t

$d(\cdot)$ = max degree of redexes

Goal: $d(t[x := u]) \leq \max(d(t), d(u), \partial(U))$

Source 2: Copies of Redexes from u

Each occurrence of x in t is replaced by a copy of u. Any redex that was already present in u is duplicated into each copy. These redexes have the same degree as in u. Bounded by $d(u)$.

E.g. $t = f\,x\,x$, $u = (\lambda y{:}\iota.\, y)\,a$. Then $t[x := u] = f\,u\,u$ contains two copies of the redex in u.

Warm-up (Lemma 1)

Bound $d(t[x := u])$

Hyp: $x : U$, substituting u for x in t

$d(\cdot)$ = max degree of redexes

Goal: $d(t[x := u]) \leq \max(d(t), d(u), \partial(U))$

Source 3: Genuinely New Redexes

Suppose x appears in t in an applied position: t contains a subterm $x\,v$. Since $x : U$, we must have $U = A \to B$ for this to type-check.

Now if u happens to be a λ-abstraction $\lambda y{:}A.\, u'$, then after substitution we get $(\lambda y{:}A.\, u')\, v$ — a new redex that existed in neither t nor u.

The degree of this new redex is $\partial(A \to B) = \partial(U)$. Bounded by $\partial(U)$.

E.g. $t = x\,v$ with $x : \iota \to \iota$, $u = \lambda y{:}\iota.\, y$. Then $t[x := u] = (\lambda y{:}\iota.\, y)\, v$ — a new redex of degree $\partial(\iota \to \iota) = \partial(U)$.

Warm-up (Lemma 1)

Bound $d(t[x := u])$

Hyp: $x : U$, substituting u for x in t

$d(\cdot) =$ max degree of redexes

Goal: $d(t[x := u]) \leq \max(d(t), d(u), \partial(U))$

Lemma 1: Conclusion

In all three cases, $\partial(\rho') \leq \max(d(t), d(u), \partial(U))$. $\square$

Take-away

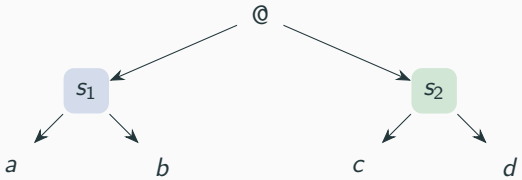
Substitution can create new redexes, but their degrees are controlled: they cannot exceed the degrees already present in t, u, or the type of the substituted variable. This analysis will be the engine behind every lemma that follows.

From Substitution to Contraction

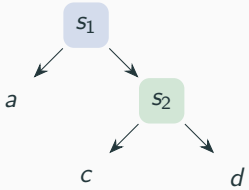
Subterm Occurrences Cannot Partially Overlap

The warm-up told us what happens to redexes *inside* a substitution. But when we contract a redex r inside a larger term t, we also need to understand what happens to redexes *elsewhere* in t. For this, we use the fact that terms are finite trees.

A subterm occurrence in t is identified by its *position*: the path from the root to the subterm's root. Two subterm occurrences are either **disjoint** or **nested** (one contains the other). There is no third possibility.



s_1 and s_2 disjoint

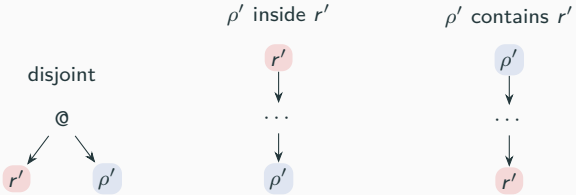


s_2 nested inside s_1

Classifying Redexes After a Contraction

We β-reduce t to t' by contracting a specific redex r inside t: the subterm r of t is replaced by the subterm r' of t', with $r \to_\beta r'$. Since $r = (\lambda x{:}A.\, M)\, N$, we have $r' = M[x := N]$ — a substitution, exactly the situation from the warm-up.

The warm-up tells us where redexes *inside* r' come from (either from M, from N, or genuinely new). But a redex ρ' in t' could also live *outside* r'. Since terms are trees, ρ' is in one of three positions relative to r':



Classifying Redexes (continued)

Let ρ' be any redex in t'. Since ρ' and r' are both subterms of the tree t', they are either disjoint or one contains the other:

1. ρ' is disjoint from r': untouched by the contraction (Case 1)
2. ρ' is inside r': lives inside the substitution $M[x := N]$ — handled by the warm-up (Case 2)
3. ρ' contains r': then $\rho' = (\lambda y{:}D.\, P')\, Q'$ with r' somewhere inside. Two subcases: (Case 3)

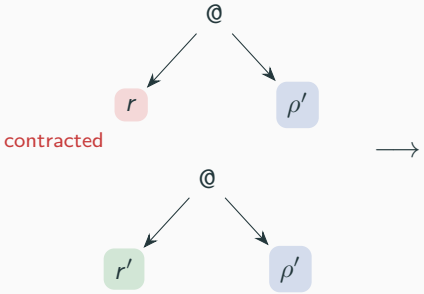
   3a r' is the entire function part $\lambda y{:}D.\, P'$ of ρ' $\Rightarrow$ **genuinely new** redex

   3b r' is strictly inside P' or Q' $\Rightarrow$ redex **already existed** in t

We now examine each case in detail.

Case 1: ρ' Unchanged

Case 1. ρ' is disjoint from r' — it was already a redex in t: same type, same degree.



Degree of ρ': unchanged, i.e. $\partial(\rho') \leq d(t)$.

Case analysis

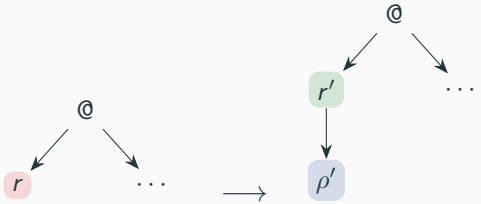
Classify all redexes of t'

Hyp: One-step β-reduction

$t \to_\beta t'$ by contracting r
$r = (\lambda x{:}A.\, M)\, N$
$r' = M[x := N] : B$
$\rho' =$ redex in t'

Case 2: ρ' Inside r'

Case 2. ρ' is inside the contractum $r' = M[x := N]$. This is exactly the situation from the warm-up: r' is a substitution, so the three sources (old redexes of M, copies of redexes from N, genuinely new) all apply.



Degree of ρ': bounded by the warm-up (Lemma 1) — $\partial(\rho') \leq d(r') \leq \max(d(M), d(N), \partial(A))$.

Case analysis

Classify all redexes of t'

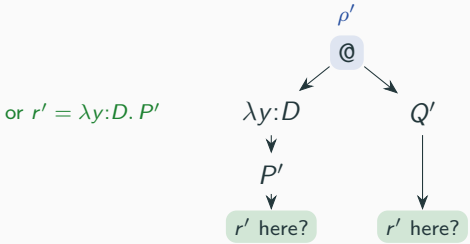
Hyp: One-step β-reduction

$t \to_\beta t'$ by contracting r
$r = (\lambda x{:}A.\, M)\, N$
$r' = M[x := N] : B$
$\rho' =$ redex in t'

Case 3: ρ' Contains r'

Case 3. ρ' is a redex that *contains* the contractum r'. Then $\rho' = (\lambda y{:}D.\, P')\, Q'$, and r' is somewhere inside. Where exactly?

Since subterms are either disjoint or nested, r' either is the function part $\lambda y{:}D.\, P'$, or is strictly inside P' or Q':



Case analysis

Classify all redexes of t'

Hyp: One-step β-reduction

$t \to_\beta t'$ by contracting r
$r = (\lambda x{:}A.\, M)\, N$
$r' = M[x := N] : B$
$\rho' =$ redex in t'

Case 3a: $\rho' = r'\ v$ (Genuinely New Redex)

$r' = \lambda y{:}D.\ P'$ (the whole function part), so $\rho' = r'\ Q'$ — a **genuinely new** redex. This is the same mechanism as Source 3 from the warm-up: a non-λ term in function position becomes a λ after reduction.

Recall that r is itself the redex being contracted, so $r = (\lambda x{:}A.\ M)\ N$ — an application, not a λ. But its type B could be an arrow type $B = D \to E$ (because M could return a function). So $r\ v$ in t is well-typed but *not* a redex, because r is an application, not a bare λ.

Case analysis

Classify all redexes of t'

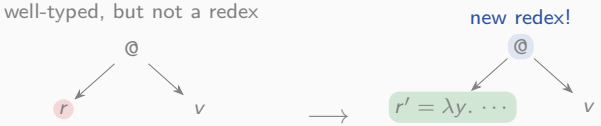
Hyp: One-step β-reduction

$t \to_\beta t'$ by contracting r
$r = (\lambda x{:}A.\ M)\ N$
$r' = M[x := N] : B$
$\rho' =$ redex in t'

Case 3a (continued)

Degree of ρ': since $r' : B$ and B is an arrow type, $\partial(\rho') = \partial(B) < \partial(A \to B) = \partial(r)$.

{**Example:** $r = (\lambda x{:}\iota.\, \lambda y{:}\iota.\, y)\, a$, $v = b$. The term $r\, v = ((\lambda x{:}\iota.\, \lambda y{:}\iota.\, y)\, a)\, b$ — the outer application is not a redex because its function part is $(\lambda x{:}\iota.\, \lambda y{:}\iota.\, y)\, a$, an application. After contracting the inner redex, we get $(\lambda y{:}\iota.\, y)\, b$, which is now a redex.}



Case analysis

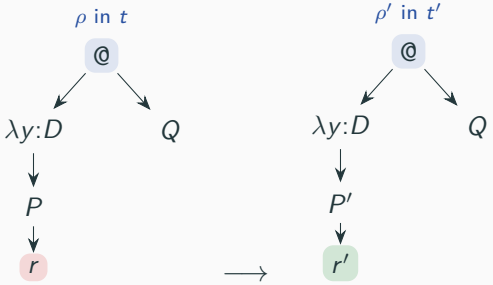
Classify all redexes of t'

Hyp: One-step β-reduction

$t \to_\beta t'$ by contracting r
$r = (\lambda x{:}A.\, M)\, N$
$r' = M[x := N] : B$
$\rho' =$ redex in t'

Case 3b: r' Strictly Inside a Subterm

r' is strictly inside P' or Q'. The redex $\rho = (\lambda y{:}D.\, P)\, Q$ already existed in t, with r in place of r'.



Degree of ρ': same as ρ in t (r and r' have the same type).

Case analysis

Classify all redexes of t'

Hyp: One-step β-reduction

$t \to_\beta t'$ by contracting r
$r = (\lambda x{:}A.\, M)\, N$
$r' = M[x := N] : B$
$\rho' =$ redex in t'

Proof of Lemma 2 (Degree Non-Increase)

Lemma 2

Lemma 2 (restated)

If $t \to u$ then $d(u) \leq d(t)$.

Proof. Assume $t \to u$ is obtained by contracting a redex $r = (\lambda x{:}A.\, M)\, N$ in t to $r' = M[x := N]$, so that r is a subterm of t and r' is the corresponding subterm of u. We classify each redex ρ' in u by its position relative to r', using the four cases from the previous analysis. In each case we show $\partial(\rho') \leq d(t)$.

Proving Lemma 2

Show $d(u) \leq d(t)$

Hyp: $t \to u$ by one-step β-reduction

$t \to_\beta u$ by contracting r

$r = (\lambda x{:}A.\, M)\, N$

$r' = M[x := N] : B$

$\rho' =$ redex in u

Lemma 2 (continued)

Case 1 (ρ' disjoint from r' — unchanged): $\partial(\rho') = \partial(\rho) \leq d(t) \Rightarrow$ **not new**.

Case 2 (ρ' inside r'): By the warm-up (Lemma 1), $\partial(\rho') \leq \max(d(M), d(N), \partial(A))$, all $\leq d(t) \Rightarrow$ **possibly new but bounded**.

Case 3a ($\rho' = r'\,v$ — genuinely new): $\partial(\rho') = \partial(B) < \partial(A \to B) = \partial(r) \leq d(t) \Rightarrow$ **possibly new but bounded**.

Case 3b (r' strictly inside ρ' — already existed): $\partial(\rho') = \partial(\rho)$ (r and r' have the same type), so $\leq d(t) \Rightarrow$ **not new**.

All redexes in u have degree $\leq d(t)$. $\square$

Proving Lemma 2

Show $d(u) \leq d(t)$

Hyp: $t \to u$ by one-step β-reduction

$t \to_\beta u$ by contracting r
$r = (\lambda x{:}A.\, M)\, N$
$r' = M[x := N] : B$
$\rho' =$ redex in u

Proof of Lemma 3 (Key Lemma)

Lemma 3

Lemma 3 (restated)

Let r be a redex of maximal degree n in t, with all redexes strictly inside r having degree $< n$. Then contracting r strictly reduces the count of degree-n redexes.

Proof. Let $r = (\lambda x{:}A.\, M)\, N$ with $\partial(r) = \partial(A \to B) = n$. Contracting gives $r' = M[x := N] : B$.

The contracted redex r is destroyed. It remains to show: no new degree-n redex appears. We again classify each ρ' in t' by position relative to r', but now the maximality hypothesis gives us *strict* inequalities.

Proving Lemma 3

Fewer degree-n redexes in t'

Hyp: $\partial(r) = n$ is maximal in t

All redexes inside r: degree $< n$

$t \to_\beta t'$ by contracting r

$r = (\lambda x{:}A.\, M)\, N$

$r' = M[x := N] : B$

$\rho' =$ redex in t'

Lemma 3 (continued)

Case 1 (ρ' disjoint from r' — unchanged): same degree as in $t \Rightarrow$ **not new**.

Case 2 (ρ' inside r'): By the warm-up (Lemma 1), $\partial(\rho') \leq \max(d(M), d(N), \partial(A))$. Since r is innermost of maximal degree, all redexes inside M and N have degree $< n$, and $\partial(A) < \partial(A \to B) = n$. So $\partial(\rho') < n \Rightarrow$ **below threshold**.

Proving Lemma 3

Fewer degree-n redexes in t'

Hyp: $\partial(r) = n$ is maximal in t

All redexes inside r: degree $< n$

$t \to_\beta t'$ by contracting r

$r = (\lambda x{:}A.\, M)\, N$

$r' = M[x := N] : B$

$\rho' =$ redex in t'

Lemma 3 (continued)

Case 3a ($\rho' = r'\,v$ — genuinely new): $\partial(\rho') = \partial(B) < \partial(A \to B) = n \Rightarrow$ **below threshold**.

Case 3b (r' strictly inside ρ' — already existed): same type and degree as ρ in t (r and r' have the same type) $\Rightarrow$ **not new**.

In every case, no new degree-n redex appears, while r has been destroyed. $\square$

Proving Lemma 3

Fewer degree-n redexes in t'

Hyp: $\partial(r) = n$ is maximal in t

All redexes inside r: degree $< n$

$t \to_\beta t'$ by contracting r

$r = (\lambda x{:}A.\, M)\, N$

$r' = M[x := N] : B$

$\rho' =$ redex in t'

Summary

Summary of the Proof

Lemma 1 (Substitution): $d(t[x := u]) \leq \max(d(t), d(u), \partial(U))$.

Lemma 2 (Non-Increase): Contracting any redex does not increase the degree.

Lemma 3 (Key): Contracting an innermost maximal-degree redex strictly reduces the count of maximal-degree redexes.

Theorem: The measure $\mu(t) = (d(t), \nu(t))$ strictly decreases at each step. Well-foundedness of $<_{\text{lex}}$ gives termination.

The critical inequality:

$$\partial(A) < \partial(A \to B)$$

Without the $+1$ in $\partial(A \to B) = \max(\partial(A), \partial(B)) + 1$,

the substitution lemma would fail, and the proof would collapse.

This $+1$ is the *entire reason* that STLC normalizes.